NC STATE UNIVERSITY

Introduction to ns-3

- Summer Camp 2026

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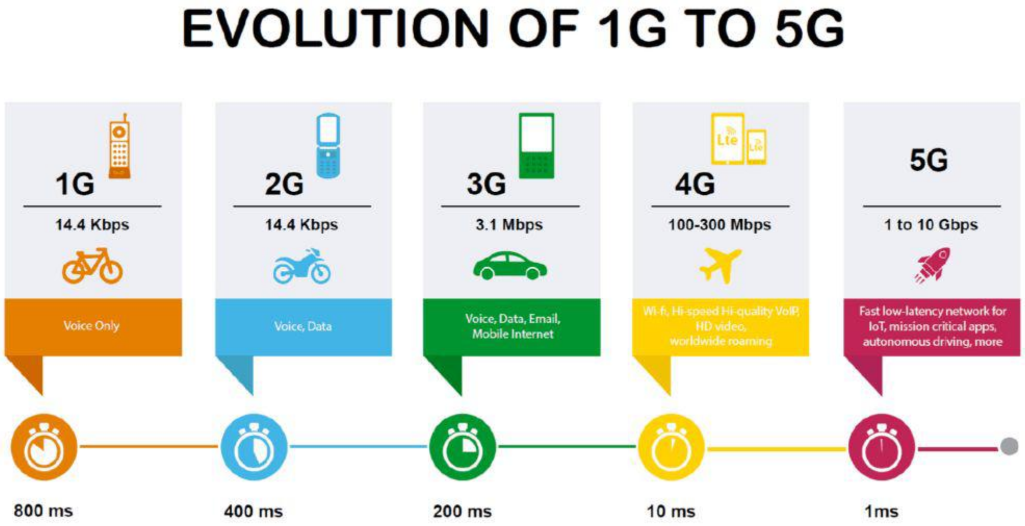
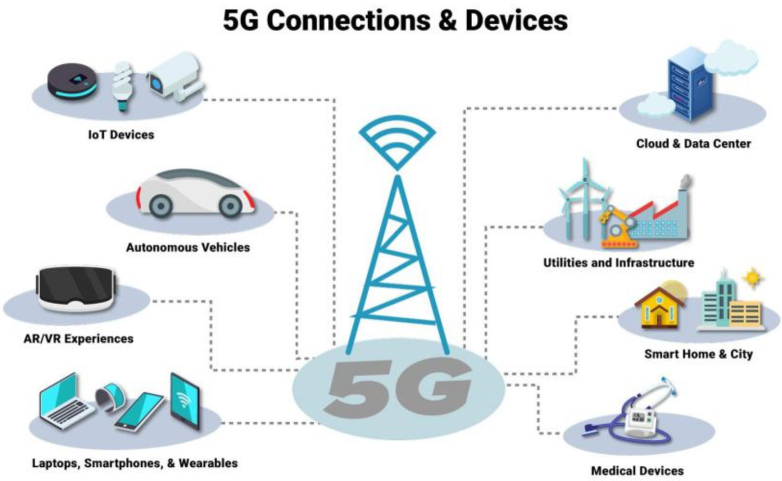
Roadmap

1. **Modern network systems:** Cellular networks, Open RAN, vehicular networks, and satellite networks
2. **Understand why simulation is needed**
3. **Introduce ns-3 architecture**
4. **Install ns-3 and examples**

Cellular Networks



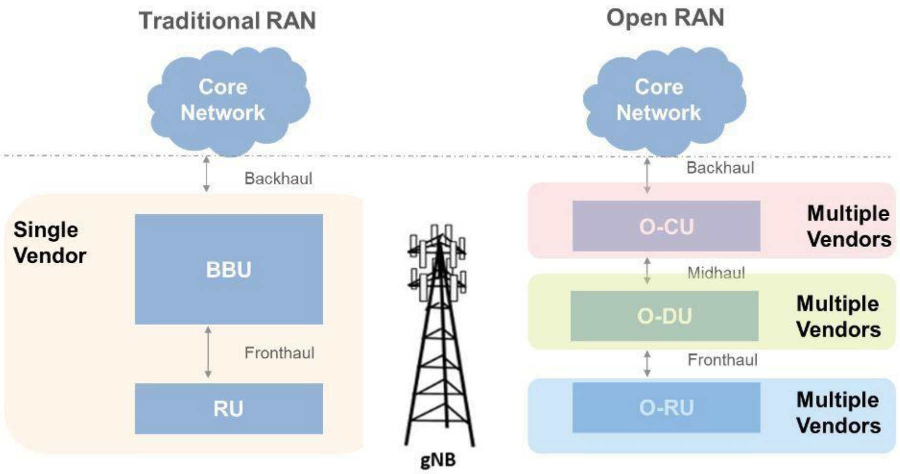


Traditional vs. Open RAN

RAN (Radio Access Network): the **base-station side** of a cellular network. It connects user devices to the cellular core network.

Open RAN splits the base station into open components:

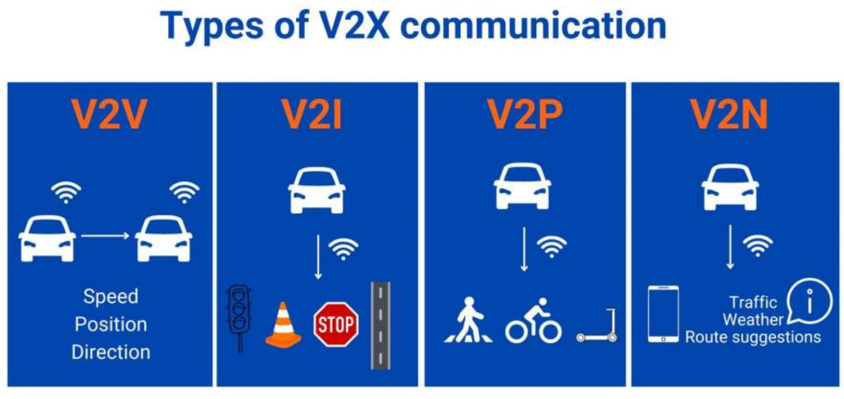
- **O-RU**: Open Radio Unit
- **O-DU**: Open Distributed Unit
- **O-CU**: Open Centralized Unit

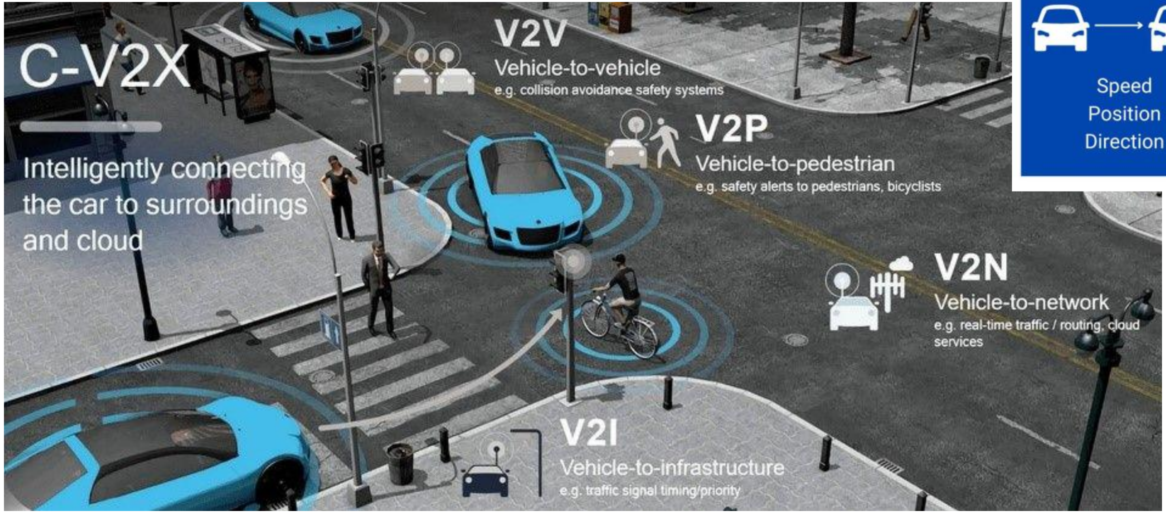


Traditional RAN: one vendor provides a closed base-station system.
Open RAN: different vendors can provide different RAN components through open interfaces.

Vehicular Networks

Vehicles are connected via wireless networks and coordinated by a few cloud servers.

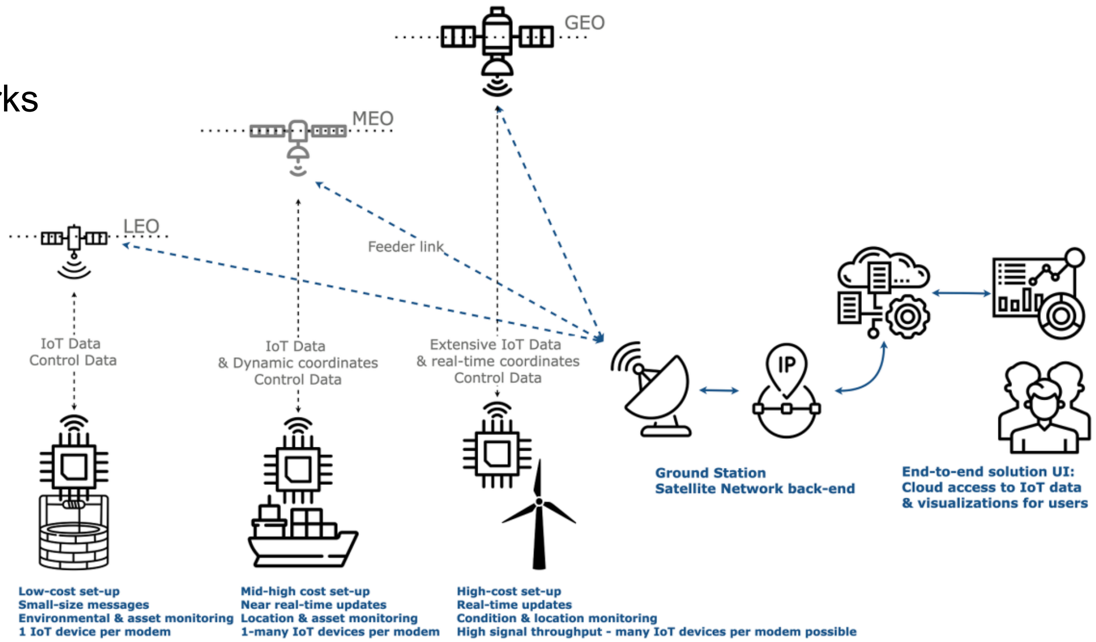




Satellite Networks

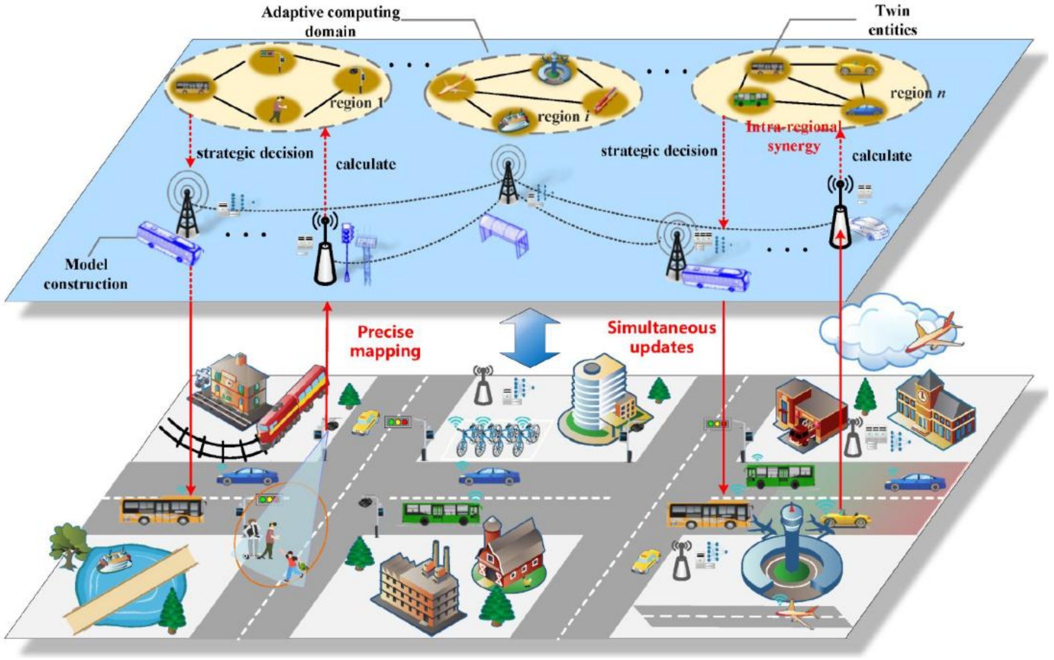
Satellite networks provide connectivity for remote IoT devices where terrestrial networks are unavailable.

- Low-Earth Orbits (LEO)
- Medium Earth Orbit (MEO)
- Geostationary Orbit (GEO)



Real Testbed vs. Simulation Platform

- Testing and validations on real testbed may not be cost-effective.
- More factors, such as safety risks and scalability, need to be considered in real testbeds.



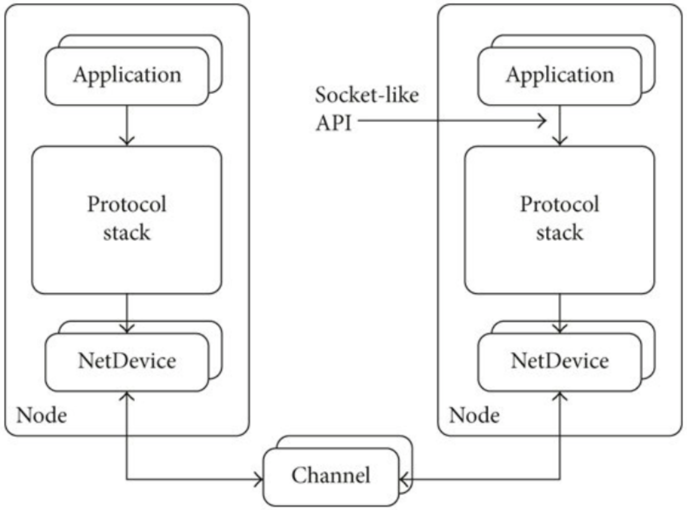
What is ns-3?

- ns-3 is a discrete-event network simulator
- It is widely used in networking research and education
- It models packets, links, devices, protocols, and applications
- Users write simulation programs, run experiments, and collect results
- ns-3 is not a real network, but it can model many network behaviors

ns-3
Network Simulator

A Network in ns-3: From Real Components to Software Objects

1. **Node**: A container representing a computer, router, phone, sensor, or access point
2. **Application**: The traffic source or receiver, such as a client or server.
3. **Protocol Stack**: Handles networking protocols such as IP, TCP, UDP, and routing
4. **NetDevice**: A network interface used to send and receive packets, such as a Wi-Fi card or Ethernet card
5. **Channel**: The communication medium connecting NetDevices across Nodes



What Can We Measure?

Bandwidth usage: Exceeding bandwidth limit slows traffic, and causes network congestion.

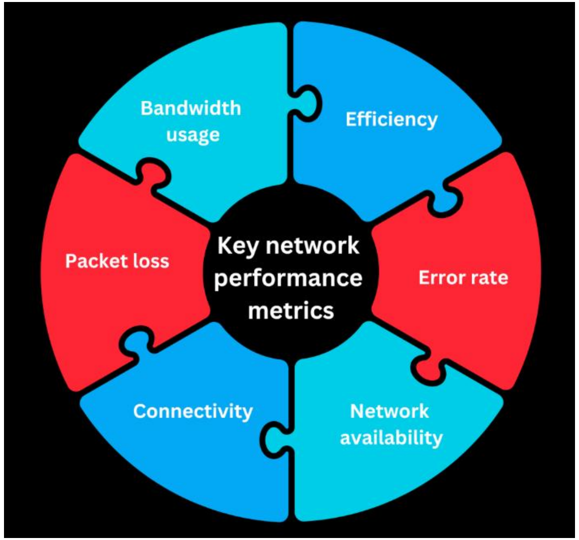
Packet loss: Data packets lost during the journey between two destinations, and can adversely affect user services.

Efficiency: Measures the real amount of data packets sent over the network and detects areas with throughput and re-transmission.

Network availability: Enabling you to observe the uptime of your network within a certain period.

Connection: This metric allows you to check the connectivity between devices and nodes in your network.

Error rate: The percentage of requests that didn't receive a response or failed.



Installing ns-3

- ns-3 Installation Guide：
  https://www.nsnam.org/docs/release/3.38/installation/singlehtml/index.html
- Use Linux or Ubuntu environment
- Windows: VMware Workstation Player and download Ubuntu 20.04 ISO

The First ns-3 Example

- first.cc is the "Hello World" example of ns-3
- It creates two network nodes
- The two nodes are connected by a point-to-point link
- One node runs a UDP Echo Server
- The other node runs a UDP Echo Client
- The client sends one packet, and the server sends it back

```
//        10.1.1.0
// n0 -------------- n1
//    point-to-point
```

https://www.nsnam.org/doxygen/dir_c1b9e66bce6b1b2db4a457a593cee634.html

Takeaways

1. Modern networks are everywhere: cellular, Open RAN, vehicular, and satellite networks.
2. Real network experiments can be expensive, risky, and hard to repeat.
3. ns-3 provides a safe and repeatable way to simulate and evaluate network systems.
4. In ns-3, real network components are modeled as Nodes, Applications, NetDevices, Channels, and Protocol Stacks.
5. Finally, we walked through how to install ns-3 and run basic examples.